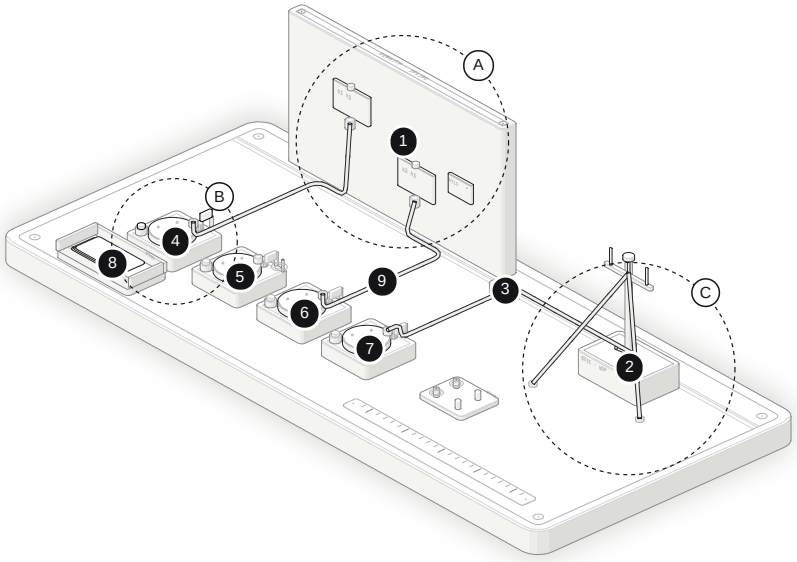
loxide



How does a request wait for the network without a single syscall?

One reactor thread per core, each with its own io_uring ring of 8192 entries. Connections arrive on trunk ports — TCP with TLS and QUIC over UDP for HTTP/3 — and every reactor serves any of them. Each posted ticket is a submission; each lit one, a completion.

1 ring
IO_URING INSTANCE PER CORE

8192 entries
SUBMISSIONS AND COMPLETIONS PER RING

4096 slots
RECEIVE BUFFERS PER REACTOR, 32 KIB EACH

0 syscalls
SPENT WAITING FOR WORK TO ARRIVE

3 protocols
HTTP/1.1, HTTP/2 AND HTTP/3 ON ONE HOST

PARTS

01 **Trunk panel**
Two listener ports, plain and TLS; the small block is kernel TLS.

02 **Radio mast**
QUIC over UDP — ngtcp2 and nghttp3 feed HTTP/3 in from the air.

03 **Distribution block**
Where cords meet: the junction the completions call back through.

04 **Operator 1**
Reactor on core 1, patching calls on its own ring.

05 **Operator 2**
Reactor on core 2; its cord rests on the hook between calls.

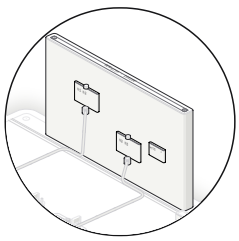
06 **Operator 3**
Reactor on core 3, fed from the second trunk port.

07 **Operator 4**
Reactor on core 4, fed through the distribution block.

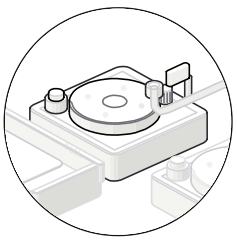
08 **Request pool**
Request objects dealt from a tray, not allocated on the call path.

09 **Patch cord**
One cord per live connection, port to operator.

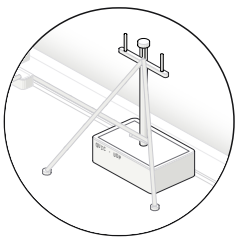
CLOSE-UPS



(A) **Trunk panel**
ALPN negotiates h2 on the TLS port; without it the driver peeks for the HTTP/2 preface and falls through to HTTP/1.1. Kernel TLS moves cipher work to the NIC.

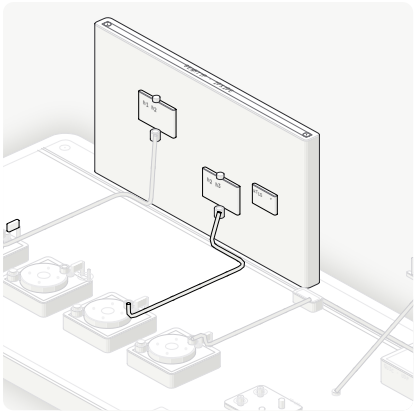


(B) **A patch ring**
Twelve slots around one ring — the ring's entries. The lit dot is the completion being reaped right now.



(C) **The mast**
QUIC needs its own timeouts, datagram routing and peer-migration pinning, so the mast is guyed like the real thing.

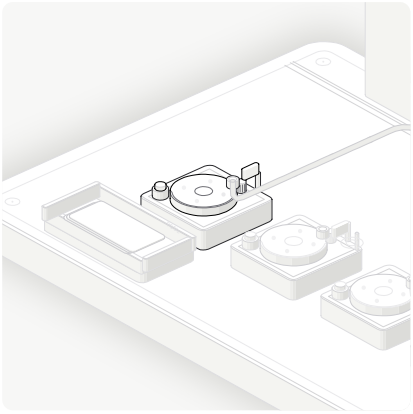
How it works



01

The call lands

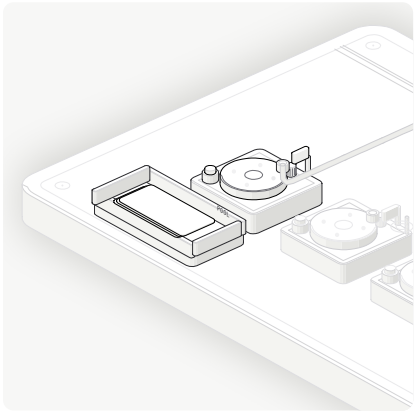
A connection arrives on a trunk port. The kernel accepted it already; the trunk cord is waiting at the port plate.



02

A reactor picks up

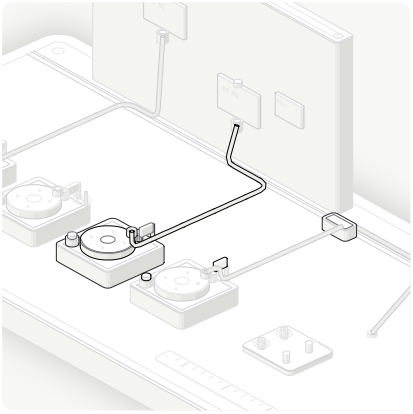
Operator 1's lamp lights and the duty flag goes up: this call belongs to reactor 1 until it is answered.



03

The ring turns

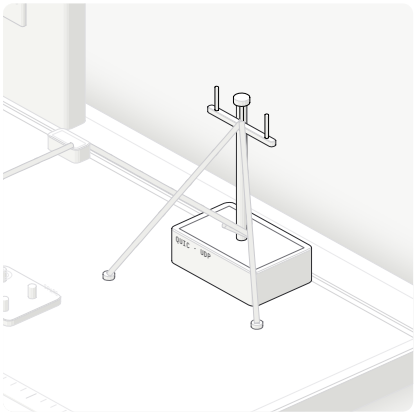
Each completion advances the patch ring one slot. The readout counts completions — work reaped without waiting on a syscall.



04

Hand-off, not migration

Another port, another operator: every core's ring runs on its own pinned thread, so a call hands off cleanly — through the distribution block, same four-method handler at the end.



05

Radio, no wires

QUIC skips the trunk panel entirely: the mast feeds HTTP/3 streams straight into the exchange, one handler call per stream.

WHY A RING PER CORE?

epoll asks the kernel what is ready, then reads each socket. io_uring is two shared ring buffers: submissions go in one side and completions appear on the other without a syscall per event. One ring per core with DEFER_TASKRUN means a request lives its whole life on one core.

SPECIFICATION

Engine	Ioxide, NuGet 0.15.251	TLS	OpenSSL, ALPN, optional kernel TLS
Reactors	one pinned thread per core	HTTP/3	ngtcp2 + nghttp3 over QUIC
Ring	8192 entries, DEFER_TASKRUN	Parser	Glyph11, hardened HTTP/1.1 + h2
Receive	4096 slots × 32 KiB, pre-allocated		